Cu-insulin Incubation study

From raw ABF recordings to reviewed events, analysis and publication figures

Four primary recordings + one excerpt · Schmitt detector · 35 reference figures

Two ways to use this manual

Reproduce the reference. Enter the specified settings and retain the automatic acceptance decisions. Compare your saved checkpoints with the reference counts.

Develop a curated analysis. Inspect individual events and whole clusters, choose which events to keep, confirm the selection, and rerun the dependent analyses before producing final figures.

Edition: 29 September 2026

Target app: NanoScope 40.15.59, unified incubation study

20260929T215919236142Z-nogit-7a81ac428875

Open Analysis → Incubation study… and choose the study profile. Complete each operation through the saved study stages and curate events in the main app.

Find your next step

Before you start: verify the app build 3

One incubation entry for current and future studies 3

1. Choose your reproduction route 4

2. Add the exact source files 4

3. Prepare each recording 6

4. Detect, measure and run null controls 7

5. Review individual events and approve the selection 9

6. Optionally select whole clusters before final analyses 10

7. Compute the final recording comparisons 11

8. Extract the final features and fit populations 12

9. Generate all 35 final figures 13

10. Style, export and compose the figures 14

11. Save, reopen and verify the completed workspace 15

Final checklist and troubleshooting 16

Workflow order

Sources → preparation → detection → individual review → confirm selection → preliminary clustering → optional cluster review → confirm again → final comparisons and fits → figures → styling and export → save/reopen.

A cluster review needs an initial clustering result. After changing the selected events, generate a new analysis and report from that selection.

This manual takes you from the five source ABF recordings to reviewed detections, recording comparisons, waveform clustering, the 30-hour population appendix and 35 saved figures. You choose and run each operation in NanoScope's **incubation study workspace**. No notebook execution is required. The current Cu reference uses the model-free **Schmitt v4** detector; leave the trained-model controls unused for this study.

This edition describes NanoScope **40.15.59**, build **20260929T215919236142Z-nogit-7a81ac428875**, installed in Applications. It includes the shared study launcher, main-workspace curation and explicit review confirmation before dependent analyses.

Before you start: verify the app build

1. Open the installed `/Applications/NanoScope.app` and choose **Help → About NanoScope…**. This edition targets build **40.15.59**, full build ID **20260929T215919236142Z-nogit-7a81ac428875**. The marketing version `0.1.0` alone does not identify the build.
2. The current study entry is **Analysis → Incubation study…**. The original fixed Zn notebook recipe remains within that launcher as **Advanced: original Zn reference recipe…**; its automated run does not consume the manual study's curated snapshots.
3. Create a separate empty workspace folder for each study. Preserve a complete copy of any reference workspace before developing an alternative curation, so that both selections remain available for comparison.
4. Allow each operation to finish before starting the next. The controls on the left scroll vertically; drag the central divider if labels or fields are cramped. Use **Methods / history** to inspect settings that produced completed results.

Already have the saved study? Choose **Analysis → Incubation study… → Open saved study…** to inspect it. This restores saved results; it is not a fresh reproduction. Use the profile's **Create study…** with a new empty folder and the original ABFs to repeat the analysis.

One incubation entry for current and future studies

Open **Analysis → Incubation study…**. For this reference choose **Cu incubation**, enter **Study name** and optionally **Sample**, then click **Create study…** and select a new, empty folder. That action creates the workspace and opens its stages; no second New workspace operation is needed. To restore a completed study, use **Open saved study…** and select its whole workspace folder. To resume one already open, select it in the open-study list and click **Continue selected study**. Opening another study preserves the other open studies and their drafts. Use one editing session per folder.

General, Zn and Cu share **1 Sources**, **2 Prepare**, **3 Detect / review**, **4 Compare**, **5 Features / populations** and **6 Figures**, with curation in the main app's Event table, Gallery and Inspector. The Zn profile starts with reflected median edges and Cu starts with shrinking edges. A profile supplies starting settings; enter every reference parameter below rather than assuming the profile has set them all. Opening an existing study restores its scientific method and saved settings without applying new-profile defaults.

For future samples or conditions choose **General incubation**, give the study an explicit name and sample, and enter the actual recording metadata. On **1 Sources**, edit **Study name**, **Sample / material** and **Condition / treatment label**, then **Apply study details**. A trained model is optional: prepared data

can use signal detection or the main workspace's compatible detectors. Changing the sample does not validate the Zn or Cu scientific parameters for it. The Acid profile uses the same launcher and main curation tools while preserving its calibrated CUSUM and family-analysis method.

This is an analysis protocol for acquired recordings. The supplied files and metadata do not specify a complete laboratory incubation recipe: temperature, pH, metal concentration, preparation and agitation must come from the experimental record. Do not infer them from this interface or substitute the recording concentration for an undocumented incubation concentration.

1. Choose your reproduction route

Reference reproduction: enter the settings in this manual and retain the detector's automatic acceptance decisions. The checkpoints below describe the delivered reference study. You still confirm the event selection before downstream analysis.

Your curated study: inspect and change individual events, optionally use a preliminary clustering to accept or reject whole groups, then recompute the final analyses from that selection. Your final counts and figures should reflect your decisions and may differ from the reference. Record the reason for each scientific selection in your study notes.

The full sequence is **raw files → preparation → detection and null controls → initial event review and confirmation → preliminary features/clustering if you want cluster curation → final review confirmation → final comparisons and features/populations → figures and styling → save and reopen**. Preliminary clustering exists to help choose events; it is not the final population result after you change the selection. The 200 mixture bootstrap fits later in this manual estimate fit variability; they are unrelated to the number of times you inspect or revise your selection.

To inspect the previously completed reference, choose **Analysis → Incubation study… → Open saved study…** and select `/Users/abrahamcampos/Documents/Codex/2026-09-12/le/outputs/cu-study/manual-study`. To reproduce independently, select **Cu incubation** in the launcher, enter the study name, use **Create study…**, and choose a new empty folder such as `Cu-incubation-reproduction`. To develop a different selection from the reference, first save and close it, duplicate the entire workspace folder in Finder, and open the duplicate. The manual workspace saves completed operations automatically.

2. Add the exact source files

In the main NanoScope window, open **Analysis → Incubation study…**. Select **Cu incubation**, enter **Study name** and optionally **Sample**, click **Create study…**, and choose a new empty output folder. In the resulting study window click **Add raw ABF…**. Add these five files. Keep their original names; the spelling `adjuested` belongs to the actual primary 30-hour filename.

Recording tag	Source location
`cu_2h`	`Desktop/Cu2Data/100 pM Cu_2h_16 nm_50 PEG_0011_-200 mV_Conductive_Good_10-15 min.abf`
`cu_10h`	`Desktop/Cu2Data/100 pM_10 h_Cu_16 nm_50k PEG_0004_-200 mV_Conductive_good.abf`

Recording tag	Source location
cu_18h	Desktop/Cu2Data/100 pM Cu_18 h_18 nm_50 PEG_0002_-200 mV_Conductive_Very Good_10-14.5 min.abf
cu_30h	Desktop/Cu_30h_100 pM_18nm_50 PEG_0001_Process_-200 mV_Conductive_Very Good_Baseline adjuested.abf
cu_30h_excerpt	Desktop/Cu2Data/Cu_30h_100 pM_18nm_50 PEG_0001_Process_-200 mV_Conductive_Very Good_10-15 min.abf

Here Desktop means `/Users/abrahamcampos/Desktop/` on the reference computer. On another computer, select your copies of the same files. The new primary 30-hour source is directly on the Desktop. The old 10-15-minute excerpt is a separate supplementary recording-state comparison and must not replace it.

Use the recording selector at the top of the study window to work on one recording at a time. In **1 Sources**, enter the following common settings for all five recordings:

Source control	Value
Concentration (pM)	100
Voltage (mV)	-200
Current channel (1-based)	1
Event orientation	**Downward in source**
ABF numeric scaling	**Notebook float32 convention**
Median window edges	**Shrinking (Cu convention)**

Then set the recording-specific values:

Control	cu_2h	cu_10h	cu_18h	cu_30h	cu_30h_excerpt
Recording tag	cu_2h	cu_10h	cu_18h	cu_30h	cu_30h_excerpt
Incubation (h)	2	10	18	30	30
Physical pore ID	cu_2h	cu_10h	cu_18h	cu_30h	cu_30h_excerpt
Pore diameter (nm)	16	16	18	18	18
Keep from source time (s)	0	0	0	300	0
Keep until source time (s; 0 = end)	0	0	0	900	0
Exclude source intervals	blank	0:20;370:380	blank	blank	blank

Control	cu_2h	cu_10h	cu_18h	cu_30h	cu_30h_excerpt
Add current channel (1-based; 0 disables)	0	0	0	2	0
Additional channel values are pA (override header)	off	off	off	on	off
Supplementary recording (exclude from pooled series)	off	off	off	off	on

Click **Apply recording details** for each recording. Do not assume that selecting another recording applies uncommitted edits. The separate primary pore IDs prevent the software from treating matching nominal diameters as proof of one shared physical pore. The supplementary excerpt is excluded from the pooled incubation series.

The 30-hour channel reconstruction matters. In the `Baseline adjuested.abf` export, channel 1 contains baseline-subtracted current and channel 2 contains the removed current baseline, although its header says mV. Adding channel 2's numeric values as pA reconstructs the raw current used by the reference notebook. This is a specific interpretation of that processed export, not a general conversion from voltage to current. Apply it only to `cu_30h`.

The keep interval includes its start and excludes its stop; the `cu_10h` excluded intervals remove both endpoints. These times are measured from the beginning of the supplied file, regardless of a `10-15 min` filename or acquisition header. All five sources are 50,000 Hz. The primary 30-hour retained interval has 599.84982 s of exposure; the last retained source sample is at 899.84980 s.

3. Prepare each recording

Select a recording, open **2 Prepare**, and enter these settings. Repeat for all five recordings.

Prepare control	Reference value
Apply sequential notch	off
Notch frequencies (Hz, comma separated)	blank
Notch Q	30; unused while notch is off
Round notch output to float32	off
Baseline method	**Event-masked median**
Mask: rough window (s)	`0.5`
Mask: baseline window (s)	`0.5`
Median-grid decimation (samples)	`10`

Prepare control	Reference value
Mask threshold (MAD sigma)	3
Mask dilation (ms)	5
Robust median / SavGol window (s)	0.5; unused for Event-masked median
SavGol polynomial order	3; unused for Event-masked median
Despike kernel (samples; 0 disables)	0

Click **Prepare selected recording** and wait for completion. Choose **Raw current**, **Filtered current and baseline**, then **Prepared current** in the preview selector and inspect the trace. **Show full recording** restores the overview. For the masked median, ensure the baseline follows the recording drift rather than the event excursions. The Cu baseline window and dilation are 0.5 s and 5 ms; those values differ from the general workspace defaults.

The baseline is calculated on the concatenated retained samples, matching the notebook. Detection and merging will not cross a gap removed in Sources. Use **Export prepared CSV and metadata…** if you want a separate preparation record. Keep the saved metadata together with the CSV.

Expected retained exposure before detection:

Recording	Retained samples	Exposure (s)
cu_2h	15,008,304	300.16608
cu_10h	18,711,198	374.22396
cu_18h	13,868,658	277.37316
cu_30h	29,992,491	599.84982
cu_30h_excerpt	15,076,224	301.52448

4. Detect, measure and run null controls

Open **3 Detect / review** for the active recording. Scroll to the signal detector controls and set all values below. Click **Detect and measure signal events**. This button performs Cu detection and measurement together. The model inference, probability decoding and model-candidate measurement controls farther down the page belong to the independent model route and are unused here.

Signal control	Value	Interpretation
Signal mean window (samples)	5	0.1 ms at 50 kHz; detection smoothing
Signal enter (MAD sigma)	4	Noise-dependent entry threshold

Signal control	Value	Interpretation
Minimum enter threshold (pA)	8	Entry uses the larger of 4 × MAD sigma and 8 pA
Signal exit (MAD sigma)	1	Lower hysteresis threshold
Signal merge gap (ms)	5	Merge nearby regions within a retained source segment
Signal minimum duration (ms)	0.2	At least 10 samples
Plateau minimum duration (ms)	1	Regions shorter than this are spikes
Plateau sustained depth (MAD sigma)	3	Required median depth for longer regions
Raw I0 rolling window (s)	5	Local raw-current reference
Raw I0 grid (samples)	50	Rolling-reference decimation

Candidate depth is measured on the unsmoothed prepared current. A short spike passes the automatic rule. A region lasting at least 1 ms passes as a plateau only if its sustained depth reaches 3 MAD sigma; otherwise its class is `shallow_long` and it starts rejected. All candidates remain available for inspection. `passes_gate` keeps the detector's automatic decision; `accepted` is the selection used by later analyses.

Set **Global-phase null seeds = 5**, **Block-phase null seeds = 3**, and **Null block duration (s) = 10**. Click **Run signal null controls**, then **Show null-control counts**. NanoScope also performs one sign-flip control, giving nine surrogate rows per recording. Thresholds are fixed from the real trace. These are diagnostic controls, not a measured false-discovery rate. Repeat detection and controls for each recording.

Reference checkpoints before any manual acceptance or boundary changes:

Recording	All candidates	Automatically accepted	Initially rejected
cu_2h	60	34	26
cu_10h	111	95	16
cu_18h	161	58	103
cu_30h	951	417	534
cu_30h_excerpt	729	274	455

The primary series has **604 accepted events**. The excerpt's 274 events are supplementary. If reference counts differ, check the 30-hour file and channel reconstruction, the 10-hour excluded intervals, shrinking median edges, preparation values, orientation and detector route before fitting populations. Counts alone are not enough: inspect representative accepted and rejected waveforms and their bounds.

5. Review individual events and approve the selection

1. Select the recording and click **Show all candidates**. Use **All candidates**, **Accepted only**, or **Rejected only** to choose the visible rows.
2. Click a row in **Results table** to inspect its full-resolution waveform in **Trace / event**. Check that the marked interval covers the event, that the depth is measured on the intended excursion and that a baseline fluctuation has not been mistaken for the event of interest.
3. Select one or several rows, then click **Accept selected** or **Reject selected**. The action applies to selected rows, not every row visible in the filter. Inspect the resulting accepted count. The automatic `passes_gate` value is retained separately.
4. To reproduce the reference, retain the starting automatic decisions. If you changed them while exploring, **Reset review to automatic gate** restores the automatic selection and clears the current human review status.
5. Click **Use reviewed events for analysis** once the selection is ready. Do this for all four primary recordings and review/confirm the excerpt separately. Confirmation is required even when you keep the automatic selection; it approves this selection for downstream analysis and does not claim that every event was individually inspected.

Explicit accept/reject actions record individual review evidence even when the value was already the same. A confirmation and an individual inspection are distinct records. Later acceptance, boundary, preparation or measurement changes require you to inspect and confirm again.

Use the main NanoScope event tools when needed

For richer trace inspection or boundary editing, click **Curate in main workspace…**. NanoScope opens a prepared recording copy with its events. Use the main event table, Inspector and Event gallery to inspect and curate. In the main study bar click **Save curation to study**, wait for a successful save, then **Return to study**. This captures the current event table and edited bounds for that exact study recording. Cu measurements are recalculated using the same Schmitt measurement policy and local raw-current reference. Then click **Use reviewed events for analysis**.

Edits made later in the main window are not automatically copied back to the study; click **Save curation to study** again. Confirm that the active study recording and the native review recording correspond before importing. A main-workspace import retires prior cluster memberships, so rebuild any preliminary grouping that depends on those events.

Use **Export candidate and accepted tables…** to save your decision tables. `accepted` describes the selected study set, `passes_gate` describes automatic classification, and `event_review` records explicit review actions in the current workflow.

Exact main-window event review controls

After **Curate in main workspace…**, keep the matching recording and detection table active. The persistent **Incubation study** bar identifies the exact study recording. Choose **Analysis → Event inspector…**. Use **◀ Prev** and **Next ▶** to navigate, then **Accept** or **Reject** for the current event; these actions do not advance automatically. The **Accepted** checkbox reflects its state. The A/R shortcuts perform the same decisions, and the arrow keys navigate.

For a boundary correction, drag the event mask edge in Event Inspector and wait for the edit and measurement to finish. **Revert mask**, when available, restores the recorded original bounds. The highlighted interval in the study's own trace preview is for inspection and cannot be dragged. The main

Event table, Gallery and Inspector also provide their ordinary labeling, missed-event, merge and Undo tools; Reconciled tables retain their source-editing restrictions.

Click **Save curation to study** in the main study bar and wait for successful completion, then **Return to study**. Inspect the imported bounds, measurements and decisions, then click **Use reviewed events for analysis**. Saving a main project preserves that workspace but does not commit event changes to the study. Later edits require **Save curation to study** again. Reentering compatible curation retains unsaved main-workspace edits; if upstream study data or settings changed, return to the study and open its current events because an outdated copy is refused.

Match the time axis when checking an event

The manual trace preview, native review copy, rate time blocks and 30-hour population-onset figure use retained acquisition time. Exported `original_onset_s` and `original_end_s` map to the source file before the cuts. For `cu_30h`, add 300 s to a retained onset to find it in the source. For `cu_10h`, use the exported original-time columns because the two exclusions make one fixed offset incorrect across the entire recording. Cu signal overview figures use original source time where labeled. Check both the event ID and the appropriate clock when comparing a trace with a table.

6. Optionally select whole clusters before final analyses

Skip the rejection steps in this section to retain the reference selection. To curate by morphology, first run an exploratory grouping of the initially approved events. Final comparison plots can wait until you finish this branch.

1. In **5 Features / populations**, click **Extract reviewed-event waveform features**. This reads the accepted primary events; the supplementary excerpt stays outside the pooled set.
2. Set the 12 feature columns and pooled settings in Section 8, then click **Run pooled exploratory clustering**. New runs automatically load their memberships into **Review whole clusters**.
3. In **Review whole clusters**, choose **Pooled clustering** as the source and **K-means (3 clusters)** or **HDBSCAN** as the assignment method. For an older saved analysis, click **Load saved clusters for review** first. Read **Applies to:** and check the listed recording tags; pooled review covers that snapshot's primary recordings.
4. Select one or several cluster rows. Click **Show events in selected clusters**, then inspect the individual events and source traces. Check whether the group represents the morphology you intend to keep before changing it.
5. Click **Accept selected clusters** or **Reject selected clusters** to change every event in the selected groups. **Keep only selected clusters** accepts those groups and rejects all other events within the displayed recording domain, including unassigned events unless you select that bucket too. Recordings outside the domain remain unaffected.
6. Refine individual decisions in **3 Detect / review** if needed. Cluster and individual controls edit the same `accepted` value; the last action affecting an event determines its selection.
7. Click **Use cluster-reviewed events for analysis** to confirm all recordings covered by the displayed snapshot. Confirm any other included recording separately if it remains pending. Then perform the final comparisons and re-extract features, rerun clustering and fit the desired mixture using Sections 7-8. Generate final figures only after these operations.

Noise is not an automatic rejection rule. The reference Cu pooled HDBSCAN run labels all 604 primary events as **Noise (-1)**. Rejecting that row would exclude the entire primary feature set. K-means still partitions the same observations into three groups. **Unassigned / no waveform features** is a different bucket: these events were not assigned by that saved result, including candidates outside its feature set. Neither label determines whether an event is scientifically acceptable.

You can also review **Selected-recording mixture → Mixture populations** after a preliminary mixture fit using Section 8. Check the displayed target; a `cu_30h` mixture review affects only that recording. Apply the same inspect → decide → confirm → recompute sequence. The 244-event floor still applies after your selection changes.

Acceptance changes clear dependent results but preserve the old membership snapshot for further review, so you can accept a previously rejected group again. New detections, remeasurement or a native import retire old memberships. A new clustering or mixture fit replaces its source's memberships. Cluster numbers are specific to a saved run; after refitting they can describe different events. Recheck the group's meaning and plot colors after every refit.

7. Compute the final recording comparisons

Finish preparation, detection and measurement for all five added recordings, including the supplementary excerpt. When the final primary selections are confirmed, open **4 Compare**. Set **Time blocks = 20**, **Bootstrap draws = 4000**, and **Random seed = 0**. Leave the other visible fields at the saved reference values: **Fractional floor: start = 0.02**, **stop = 0.2**, **step = 0.01**, and **Sensitivity block counts = 10,20,40**. These floor/sensitivity fields serve the Zn model analysis; the Cu Schmitt comparison does not use them to impose another acceptance gate.

Click **Compute recording comparisons**. Use the table selector and **Show comparison table** to examine **Summary**, **Accepted events**, **signal_types**, **nominal_pairs**, **drift**, and **monotonicity**. The single sustained-depth gate-transfer tables are not applicable to the spike/plateau selection rule. The summary contains four primary rows, not five.

Reference numerical checkpoints, before changed curation:

Primary recording	Accepted events	Rate (events/s)	Median dwell (ms)	Median local fractional change (%)
cu_2h	34	0.113271	0.53	4.0418
cu_10h	95	0.253859	0.74	1.9286
cu_18h	58	0.209105	0.60	4.8809
cu_30h	417	0.695174	0.50	8.1051

Rates use retained exposure, not time spent waiting during excluded intervals. The raw open-pore levels differ substantially across recordings; the primary 30-hour median is about 77.19 pA, while the supplementary excerpt is about 124.8 pA. The excerpt remains a recording-state comparison. One recording per timepoint and different pore conductances limit biological interpretation of incubation trends.

8. Extract the final features and fit populations

After any curation change, click **Extract reviewed-event waveform features** again in **5 Features / populations**. Re-extraction clears older clustering and mixture results; run the final fits afterward. With reference acceptance there are 604 pooled feature observations and 417 observations in the 30-hour mixture.

Use exactly these 12 feature names in this order in both feature-list fields. Paste as one comma-separated line:

```
log_dwell, depth_max_pA, depth_sust_pA, mean_pA, std_pA, min_dev_pA, skewness, kurtosis, area_pA_ms, t_peak_frac, f_sust, snr
```

The feature table can contain additional columns such as entropy. The 12-column list above defines this reproduction. Feature-stage fractional amplitude uses the whole-recording raw I0, whereas the detector/comparison `f_sust` uses the local raw I0. Identical column names across different stage tables do not erase that measurement distinction.

For final pooled exploration, set:

Pooled clustering control	Value
Pooled clustering features (comma separated)	the 12-column list above
Clustering seed	0
K-means components	3
PCA retained variance target	0.95
Maximum retained PCs	10
HDBSCAN minimum samples	10
Compute descriptive t-SNE	on

Click **Run pooled exploratory clustering**. Use **Show saved clustering assignments** and **Show waveform feature table** to inspect the committed tables. The reference retains four PCs to reach the 95% target and has no HDBSCAN density clusters. K-means produces a partition, but that is not evidence for three biological species. t-SNE is a descriptive display rather than a quantitative separation test.

For the quantitative 30-hour appendix, choose **cu_30h** in the mixture target selector and set:

Mixture control	Value
Mixture features (comma separated)	the same 12-column list
PCA dimension grid (blank = automatic)	2,3
Mixture K grid	1,2,3,4

Mixture control	Value
Cross-validation folds	5
Mixture bootstrap replicates	200
Mixture selection / fit seed	0
Mixture bootstrap seed	0
Minimum accepted events (0 disables)	244

Click **Fit selected-recording mixture**. Inspect **Assignments**, **Population quantification**, **Cross-validation grid**, **Mixture-weight bootstrap**, **BIC ladder** and **Exploratory crosstab** using the saved-mixture selector and **Show saved mixture table**. Read the fit warnings in **Methods / history**.

The 244-event requirement follows the reference's five training events per free parameter at the largest 3-PC, four-component full-covariance model with fivefold cross-validation. If a curated selection falls below 244, that reference fit is no longer applicable; do not silently lower the floor merely to recover the old result. A smaller, justified model specification is a new analysis and should be documented as such.

The reference chooses three PCs and four components, with hard assignments **236 / 110 / 36 / 35**. Both selected values reach the search-grid boundary. Mean silhouette is about 0.325, and the result includes weak-separation, temporal-instability and bootstrap-label-switching warnings. Treat the components as exploratory descriptions of the observations. Hard-assigned event fractions, fitted mixture weights and bootstrap intervals are different quantities; the result does not establish four molecular populations.

9. Generate all 35 final figures

Complete final comparisons, features, pooled clustering and the `cu_30h` mixture before this step. In **6 Figures**, choose a group and click **Generate selected figure group**. Figure generation reads saved results; it does not rerun detection or fitting.

Group in Figures	What to select	Outputs
Selected recording inspection (5 figures)	Each of the five recordings in turn	25
Recording comparisons	After final comparisons	5
Pooled exploratory clustering	After final pooled clustering with t-SNE	2
Selected-recording mixture	Saved target cu_30h with completed fit	3

For each recording, the five inspection images are trace overview, distributions, event examples, null controls and time dependence. The comparison group contains the primary summary, nominal-pore comparison, empirical distributions, event types and supplementary excerpt comparison. The pooled group contains PCA and t-SNE. The mixture group contains selection/fractions/waveforms, assigned-event examples and the assigned trace/onset figure.

The recording **event examples** image intentionally includes candidate examples from each automatic class, including excluded `shallow_long` events; its captions say accepted or excluded. Null-control plots also count candidates by class. Those diagnostic images are not the final accepted-event population. Accepted-event distributions, rates, extracted features and population fits use your selection.

The corresponding reference figure numbers are 1-5 (`cu_2h`), 6-10 (`cu_10h`), 11-15 (`cu_18h`), 16-20 (`cu_30h`), 21-25 (excerpt), 26-30 (comparisons), 31-32 (pooled) and 33-35 (mixture). Figure numbers are report order; the app identifies plots by their saved names. In the assigned 30-hour trace, upper markers locate actual minima inside assigned events and lower markers locate their onsets. Its clock is retained time: add 300 s to compare with the source ABF.

Choose individual figures in the saved-figure selector and inspect them in **Saved figure**. Use **Fit figure to view** if needed. PNG images, data sidecars and methods files are saved beneath your workspace's `runs` folders. A new upstream operation can clear the current figure list while preserving older files on disk. After changed curation, regenerate the desired groups from the current results; do not accidentally assemble a report from older run folders.

10. Style, export and compose the figures

1. In **6 Figures**, select a saved plot and click **Style selected figure…**.
2. Choose a general appearance preset, such as **Publication · light**, or set **Font**, **Background**, title/axis/tick text sizes, **Grid**, **Line width**, **Marker size**, and legend visibility/text size. Zero/default appearance values retain the original style. Appearance controls do not change numerical values, axis scales, event selections or scientific settings.
3. In **Panel titles and axis labels**, choose the panel and edit **Title**, **X label** or **Y label**. Include the correct scientific units and clock in custom axis labels. Per-panel edits are useful for multi-panel plots.
4. In **Linked series colors**, choose the series and click **Choose series color…**. Matching data, fills and legend keys recolor together. Color changes do not change cluster membership. Check the mapping again after refitting because a newly numbered cluster can represent a different group.
5. Click **Apply appearance** and inspect the preview. Click **Save styling to study** to persist it. Wait for the saved confirmation, then close the editor. Applying alone is a preview; saving retains styling on reopen and regeneration. You can use **Save appearance preset…** to reuse general appearance choices on other plots.
6. To export an individual plot, click **Export image…**, choose PNG, SVG or PDF and its export sizing, then choose a destination. Use **Save figure bundle…** to retain a frozen figure with its numerical/provenance resources.
7. For a multi-panel publication composition, click **Save to Figure Library…**, enter a meaningful name, then use **Save Project** in the main NanoScope window to preserve that library entry. Assemble saved panels with the main Figure Library composition tools and export the composition. A Figure Library entry is a frozen figure; regenerate and save a new entry when you want later study changes represented.

Saved study styling does not rerun detection, features, clustering or mixtures and preserves review approval. Custom event-gallery captions remain attached to the same event and plotted trace. If an event changes or is replaced, its panel uses an automatic caption until you style it again. Older captions saved without event identity must be re-entered in the editor. General font/color styling remains available.

Assemble a publication figure in the Figure Library

1. From each desired study figure editor, click **Save to Figure Library…** and enter a useful name. Save at least two entries. In the main window choose **File → Save Project…** to preserve them in a normal NanoScope project.
2. Choose **Analysis → Figure Library…**, then **Compose multi-panel figure…**.
3. Check **Use** for 2-12 saved figures. Set the panel labels and **Columns**. Use **Rows** / **Cols** spans, **Move up** / **Move down**, **Panel spacing** and panel-label appearance to arrange them.
4. Keep the study figures' own axes and legends unless the selected panels support compatible shared axes/legends. Study figures often already contain several panels.
5. Click **Update preview**, inspect the layout, then **Save to Figure Library**. Save the main project again. Select the composition and use **Export saved figure…** or **Save figure bundle…**.
6. To change a composition later, use **Revise composition…** and save the revision as a new figure. Library entries are frozen snapshots: subsequent study curation, analysis or styling does not update them. Save new entries and revise the composition after such changes.

Appearance edits control typography, colors, lines, markers and labels. They do not expose arbitrary scientific axis ranges, axis scales or plot-type changes. Preserve the units, time convention and meaning of every label you edit.

11. Save, reopen and verify the completed workspace

Click **Save workspace**. Close the manual study window, reopen **Analysis → Incubation study…**, click **Open saved study…** and select the whole workspace folder. Check the five recording tags, source settings, accepted counts, review status, saved comparison/feature/mixture tables, figure names and a styled figure. Reopening committed stages should not require recomputing them.

Keep the entire workspace folder together, including `workspace.json`, arrays, `runs` and its supporting files. Retain the original ABFs separately for raw-data reruns, together with this manual and any notes on your changes. Save the main NanoScope project separately if you used the Figure Library or want to keep native analysis copies. **Save workspace** and **Save Project** preserve different workspaces.

An older completed study can be viewed before confirming its saved selection. Before launching a new downstream analysis, inspect its events and use **Use reviewed events for analysis** for each included recording. Opening an older workspace does not approve the selections automatically. If a banner says settings differ from completed results, use **Restore completed settings** to return to the saved parameters, or intentionally run the changed upstream operation and then update its dependents.

The original Cu validation compared fresh NanoScope raw samples, prepared traces, inclusive event bounds, automatic classes, source timestamps and peak depths with independent execution of the current notebook definitions; all matched exactly. The older `CSVData_mm3` cache has different prepared baselines. It can have the same accepted counts while a few event boundaries differ. Use the current definitions and the reference checkpoints here when reproducing this study; do not treat an old cached image as proof that a new event table is incorrect.

This workflow covers the Cu incubation analysis and its waveform-population appendix. Teacher/student training and the separate IntraEventAnalysis encoder, regression and temporal-disruption experiments belong to other notebook workflows and are not reproduced by these steps.

Final checklist and troubleshooting

Check	What to do
Missing cluster-review or figure-styling buttons	Verify the full build ID in Help → About NanoScope… and open the verified app. Scroll the left control panel to find controls farther down the page.
A downstream button reports review pending	Inspect and confirm each included primary recording with Use reviewed events for analysis, or confirm the displayed cluster-review domain.
Parameters differ from saved results	Editing or saving a field does not recompute the stage. Use Restore completed settings, or intentionally run that stage and its dependents. A draft on a different recording can also block analysis; check each recording.
Cluster decisions changed the count	This is expected in the curated branch. Reconfirm, recompute comparisons and features/fits, then regenerate figures. Compare with the reference only when using its unchanged selection.
A rejected group seems to have disappeared	Select its original review source and method; acceptance edits retain that snapshot. Check the recording domain. After new detection, measurement or native import retires memberships, rerun clustering.
Only some figures appear in the selector	Finish all scientific operations first, then generate every final figure group. Earlier files in runs may describe older results.
Changes made in Event Inspector are absent from the study	Use Save curation to study in the main study bar, wait for success, choose Return to study, inspect, then confirm again.
Styling disappears after closing	Apply appearance previews the edit. Save styling to study commits it. Save Project separately for Figure Library entries and compositions.
Workspace reports a newer revision	Use one editing session per folder. Reopen the folder to load the newer saved revision; do not overwrite it from an older session.
A saved file is missing or changed	Restore the complete workspace from a known intact copy. Keep workspace.json, arrays, runs and supporting files together.

Before considering the reproduction complete, verify the recording identities, all parameter tables, accepted counts and review status; inspect examples against their source trace; inspect saved comparison and fit diagnostics; generate every required figure; save your styles; and reopen the workspace to confirm that the same results and figures return. Record the scientific reasons for any manual departures from the reference.